George Kaplan

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Audio software engineer with ten years of experience, the last four building and maintaining a hardware instrument in Rust on embedded Linux. Previously focused on full-stack and analytics work for signal processing, geospatial and manufacturing applications.

Proficient in English and Swedish. Open to hybrid or remote work.

Technical skills: Rust, C, Linux, DSP, real-time programming, JavaScript, React, Elixir, MATLAB

Experience

Elektron Music Machines, *Rust Audio Developer*

Gothenburg, Sweden · 2022 – present

- ▷ Built and released a new hardware instrument (*Tonverk*) on a new development platform based on Rust and embedded Linux, then shipped four major OS updates within its first year on the market.
- ▷ Designed and implemented signature features of *Tonverk* including the multi-sampled instrument format, *Subtracks* machine, *Auto Sampler*, and its track-routing system, drawing on my own experience as a musician and sound designer.
- ▷ Created audio effects for *Tonverk* such as *Degrader*, *Comb ± Filter*, and *Frequency Warper*. These were built in C as part of a portable DSP library, intended to share code with other products.
- ▷ Designed and built the platform's USB device-mode framework, as a Rust application driving the Linux gadget configfs interface, allowing the exposed functions to be switched at runtime without a reboot. This supports Elektron's switchable UAC2 / Overbridge operation modes and will be used in future products.

Fearless Solutions, *Software Engineer*

Baltimore, MD, USA · 2021 – 2022

- ▷ Took a React and AWS Lambda document management application from demo to production for the U.S. Centers for Medicare & Medicaid Services (CMS), and delivered it before the contract deadline. Added legal review and approval workflows, data migration, user management, and email notifications.
- ▷ Collaborated with security and operations teams to address findings from a pre-deployment audit. Closed every finding well in advance of the subsequent audit, and the deployment to production proceeded without incident.

Visionist Incorporated, *Software Engineer*

Columbia, MD, USA · 2018 – 2021

- ▷ Maintained and enhanced an audio processing and analytics service, with the API and administrative UI implemented using Phoenix (Elixir) and the audio DSP implemented in Rust.
- ▷ Scaled a suite of applications for real-time geographic data analysis up from prototype to production, focusing on performance and reliability.

Nu-Tek Precision Optical Corporation, *Engineer*

Aberdeen, MD, USA · 2016 – 2018

- ▷ Developed a 2D signal processing toolbox for MATLAB, for use in a specialized manufacturing environment.

Projects

- ▷ **pxsort**: Image glitch generator in Rust with a WebAssembly demo — github.com/g-s-k/pxsort
- ▷ **lox-lang**: Rust implementation of the Lox language from *Crafting Interpreters* — github.com/g-s-k/lox-lang
- ▷ **matlab-toml**: TOML serializer/deserializer for MATLAB — github.com/g-s-k/matlab-toml

Education

University of Maryland, *B.Sc. in Physical Sciences*

College Park, MD, USA · 2012 – 2016